Deep Learning

Final Project

Facial Expression Recognition on FER2013: Training Strategies, Transfer Learning, and External Validation

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1 Introduction

Emotion recognition has gained interest in the last years due to its applications in human-computer interactions. Also, automatized emotion recognition could help prevent car accidents caused by drivers falling asleep or even enable blind people's to detect other people emotions [1].

However, emotion recognition is a complex problem in computer vision, since we need to extract some visual features that need to be detected given different people, angles, illuminations, occlusions and image resolutions. Moreover, emotions can be expressed more subtly, or be similar to expressions of another emotion.

That's why deep learning has become a promising solution to all these issues, because neural architectures, with proper training, are able to learn highly complex hierarchical feature representations and adapt to all these condition changes.

However, to create a network able to detect emotions even with all these limitations, the dataset used for its training needs to include all these variations. This is the reason we will use FER2013 dataset for this project, because it is composed of low resolution grayscale images, containing different macro-emotions in different proportions and are visually variable. All these characteristics make these images really difficult to classify.

The goal of this project is to evaluate the impact of different training approaches and transfer learning techniques on the classifications of the emotions of FER2013 dataset. In this paper we will analyze different ways of dealing with the imbalance of the dataset emotion labels with class weights, weighted sampler, and focal loss. We will also analyze the performance of SGD with momentum, learning rate decay and gradient clipping. And we will test data augmentation techniques such as RandomCrop and TenCrop.

2 State of the art

Other researchers have also tried to use classic CNNs to classify the emotions of FER2013 dataset. For example, in Khaireddin et al. [2], they adopt VGGNet architecture and retune it to achieve the best possible performance. The transfer learning approach helps them obtain a great classification starting point to adapt to FER2013 images. In that article, they conclude that using SGD+momentum and learning decay to fine-tune the network provided the best accuracy result.

Also, in Helaly et al. [3], they used ResNet18 due to its high performance in recognition tasks. Again, they took advantage of the weights from the net already trained and changed its classifier layer to adapt to their interest dataset FER2013. In this case, they trained the classifier layer using Adam optimizer and achieved a F1-score of 64%.

Another interesting approach is Sufiuh et al. [4], where they test EfficientNetB0 by using two different transfer learning techniques: retraining only the classifier and retraining both the classifier and the feature-extraction layers. With this approach they reached an F1 of more than 65% for all emotion classes except for the sadness and fear ones.

3 Task Description

1. **Data Exploration:** loading FER2013, train/test split, class distribution analysis, imbalance visualization, resizing ($48\times48\rightarrow224\times224$), augmentation preview.
2. **Transfer Learning:** feature extraction and fine-tuning strategies.
3. **Training Experiments:** augmentation strategies, sampling methods, class weighting, focal loss, SGD variants, and advanced augmentation (Crop, TenCrop).
4. **Evaluation:** comparison of training setups, emotion-wise performance, model comparison, SOTA comparison, and external dataset testing.

4 Methodology

4.1 Dataset Analysis

The dataset used for this project is FER2013, which consists of 48x48 pixel grayscale images of faces (Figure 1). All images are centred and occupy more or less the same space in the image. Moreover, they are all labelled by the emotion they show, so we did not have to label them ourselves.

The dataset was load from Kaggle, it was already divided into a train and a test set by a stratified split. That means that the sets had almost the same emotion distribution, so the comparison of

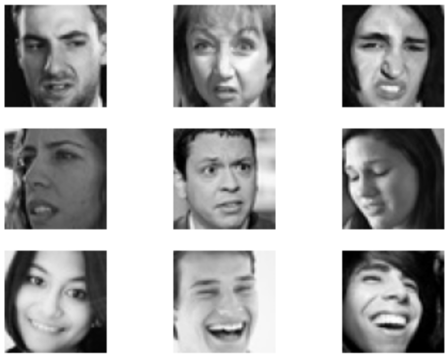

Figure 1: Visualization of some samples from FER2013 dataset

performance of the classifier on both of them would be meaningful. Then, from the training dataset, the validation set was extracted as a 10% again by a stratified split.

From the emotion distributions of the train and test datasets, we observe that the emotions classes are not equally represented in the dataset. Some emotions occur much more frequently than others resulting in a noticeable class imbalance. This is a drawback to perform deep learning tasks. This poses a challenge for deep learning models, as they may become biased toward the majority classes if the imbalance is not properly addressed. The plot from Figure 2 helps to visualize this class imbalance.

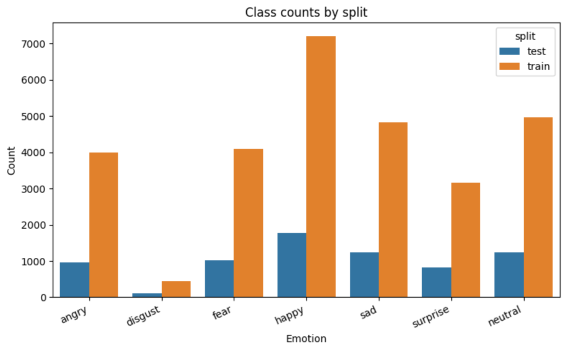

Figure 2: Bar chart showing the class imbalance in the dataset with the number of samples it contains for each emotion.

Also, some experimentations will be performed on models extracted from literature, which have an architecture prepared for pictures of 224x224 and three channels. Since the images of FER2013 are 48x48 and have only one channel, we have transformed them to have these dimensions.

Furthermore, we tried some data augmentation techniques to compensate for the small dataset. Specifically, we have used horizontal flips, small rotations and affine transformations. Since the faces from the dataset are centred, we can't perform really extreme spatial transformations, since we may lose essential information about the expressions.

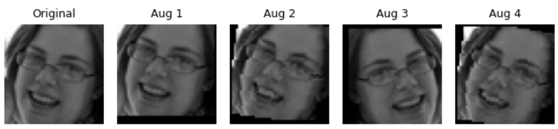

Figure 3: Example of data augmentation techniques used on our dataset

4.2 Models and Optimization

In this project, we compare the performance of three different architectures. One is a simple CNN designed by ourselves and trained from scratch using the FER2013 dataset. The other two are standard deep learning architectures, ResNet18 and EfficientNet-B0, initialized with weights pre-trained on ImageNet and then fine-tuned for facial expression recognition.

Our custom model consists of three convolutional blocks and contains 1,321,191 trainable parameters. The first two blocks have two convolutional layers to extract more complex features and each block is followed by a dropout to improve model's generalization and mitigate overfitting, which is particularly important given that FER-2013 is a noisy dataset with relatively small training size and Batch Normalization is incorporated to stabilize training and facilitate optimization. Maxpooling is applied after each convolutional block to reduce the spatial dimensions of the feature maps while retaining the most salient activations. The overall architecture of the model is summarized in Tables 1 2, and 3.

Stage	Output size
Input	48x48x1
Block1	24x24x32
Block2	12x12x64
Block 3	6x6x128
FC	1x1x7

Table 1: Custom network body architecture, describing its structure and output for each stage

Stage	Output size
Input	224x224x3
Conv/s2	112x112x64
MaxPool/s2	56x56x64
Layer1	56x56x64
Layer2	28x28x128
Layer3	14x14x256
Layer4	7x7x512
GAP	1x1x512
FC	1x1x512

Table 2: ResNet18 body architecture, describing its structure and output for each stage

Stage	Output size
Input	224x224x3
Stem Conv/s2	112x112x32
MBConv1x1	112x112x16
MBConv6x2	56x56x24
MBConv6x2	28x28x40
MBConv6x3	14x14x80
MBConv6x3	14x14x112
MBConv6x4	7x7x192
MBConv6x1	7x7x320
Head Conv	7x7x1280
GAP	1x1x1280
FC	1x1x1280

Table 3: EfficientNet-B0 body architecture, describing its structure and output for each stage

ResNet18 is a standard and widely used transfer-learning baseline for image classification. In our implementation, the model is initialized with weights pre-trained on the ImageNet dataset dataset and subsequently fine-tuned for facial expression recognition. A key feature of its architecture is residual blocks, which incorporate skip connections to facilitate gradient propagation and alleviate the vanishing gradient problem during training. Each residual block consists of two 3×3 convolutional layers, each followed by Batch Normalization, with a ReLU activation applied after the first convolution and after the residual addition. The network is organized into four residual layers, each containing two residual blocks. The overall architecture is summarized in Table 2. This network has a total of approximately 11.69M trainable parameters.

EfficientNet-B0 is another standard transfer-learning baseline for image classification. Similarly to ResNet18, it has been pretrained with ImageNet images for the classification of thousands of object classes, and its weights are publicly available. The key components of this architecture are their MBConv blocks, which use depthwise separable convolutions to make feature extraction more parameter-efficient. In our implementation, EfficientNet-B0 contains approximately 5.29 million parameters, less than half the number used by ResNet18. The overall architecture is summarized in Table 3.

5 Experiments

5.1 Baseline model

To identify the most effective configuration for our baseline model, we conducted a series of staged experiments following an ablation-study methodology. In each experiment, only one component or hyperparameter was modified while all other settings were kept unchanged. This approach allowed us to isolate the effect of each modification and assess its individual contribution to the model's performance.

First, we trained our model using data augmentation techniques. As discussed previously, data augmentation helps improve generalization and reduce overfitting, which is particularly important when working with low-resolution images such as those in the FER2013 dataset. To increase the diversity of the training data, we applied horizontal flips, small rotations, and translations throughout all experiments.

The resulting model performed noticeably better on majority classes such as happy and surprise. However, several other emotions were frequently confused, and performance on the disgust class was particularly poor. These errors may be attributed to both class imbalance and the inherent difficulty of distinguishing certain facial expressions. To address the imbalance issue, we conducted additional experiments using class weights, weighted sampling, and oversampling techniques.

We started our controlled experiments by using each of these imbalance compensating techniques. The weighted sampler was the technique that provided best results among them given its validation macro F1 scores. We thus kept this configuration for the subsequent experiments.

Our next approach was the focal loss implementation (with focal gamma=1). Focal loss places greater emphasis on difficult-to-classify samples by reducing the contribution of well-classified examples, making it a potentially suitable approach for facial expression recognition, where certain emotions are frequently confused. We therefore evaluated whether this loss function could improve the recognition of challenging emotion classes. Despite its theoretical advantages for difficult samples, focal loss did not provide a consistent improvement over cross-entropy loss in our experiments. Therefore, we keep the Cross Entropy as our loss function.

Despite the previous improvements, the model's performance had not yet reached the desired level. AdamW was used as our default optimizer due to its fast convergence and strong optimization capabilities. Nevertheless, adaptive optimizers do not always produce the best generalization performance. For this reason, we evaluated SGD with momentum, together with weight decay and learning-rate decay on plateau, to investigate whether this optimization strategy could achieve a better-performing solution.

The previous modifications led to noticeable improvements(from 0.566 to 0.627 on validation macro-f1 score), the overall performance still left room for further optimization. We therefore explored additional data augmentation techniques in an attempt to further improve the model's generalization ability. Specifically, we introduced random cropping during training and evaluated the model using TenCrop inference. This approach increases the diversity of image views presented to the model and can improve robustness to small spatial variations in facial expressions. The resulting performance showed a modest improvement over the previous configuration. At the end of the 60-epoch training schedule, the best checkpoint was still obtained at the final epoch, while the validation loss continued to improve. This suggested that the model had not yet converged. We therefore increased the training duration to 80 epochs, which led to a final macro F1 score of 0.662.

Even after extending the training duration, the model did not exhibit clear signs of overfitting. This suggested that the network capacity might still be limiting performance. Therefore, we investigated a larger architecture by adding an additional convolutional block and expanding the classifier head. However, this modification did not lead to a meaningful improvement in validation Macro F1 score. The results indicate that increasing model capacity alone was insufficient to further improve performance on the FER2013 dataset.

5.2 ResNet18 and EfficientNet-B0

For the pre-trained architectures, we followed a similar experimental pipeline, with one important difference: transfer learning was applied. Instead of training the networks from scratch, we initialized them with weights pre-trained on the ImageNet dataset. These pre-trained weights provide general visual feature representations learned from millions of images, which can then be adapted to the facial expression recognition task through fine-tuning on FER2013.

However, the effectiveness of transfer learning depends strongly on the extent of fine-tuning applied to the pre-trained network. To identify the most suitable configuration for FER2013, we performed a series of controlled experiments in which progressively larger portions of the backbone were unfrozen. By comparing the resulting performance, we determined the fine-tuning strategy that best balanced adaptation to the target task and preservation of the pre-trained representations.

For ResNet18, we explored several transfer-learning configurations by progressively unfreezing larger portions of the pre-trained backbone and modifying the classifier head. Starting from a classifier-only configuration, additional residual layers were unfrozen one at a time, and we also evaluated a deeper classifier architecture by adding an extra fully connected layer. Each configuration was assessed using validation Macro F1 score, validation loss, and per-class performance metrics. By jointly considering these criteria, we identified the combination of backbone fine-tuning and classifier design that provided the best overall performance on FER2013. The best-performing configuration was then selected for subsequent experiments.

A similar fine-tuning strategy was adopted for EfficientNet-B0. Additional MBConv stages were progressively unfrozen and compared using the same evaluation criteria as for ResNet18. The best-performing configuration was then selected for the final evaluation.

After selecting the optimal fine-tuning configuration for each architecture, the optimization pipeline developed for the baseline model was applied to both transfer-learning models.

5.3 Training configuration

Except for the components explicitly varied in each experiment, the remaining training parameters were kept constant, as summarized in Table 4.

Hyperparameter	Value
Random horizontal flip p	0.5
Random rotation degrees	10°
Random affine degrees	0°
Random affine translation	$(0.08, 0.08)$
Learning rate	0.001
Default Optimizer	AdamW
Default Loss function	Cross-Entropy

Table 4: Default hyperparameters used across the training experiments

The advanced SGD-based configurations used momentum 0.9, gradient clipping of 0.1, and ReduceLROnPlateau scheduling, while focal-loss experiments used $\gamma = 1$.

6 Results

6.1 Baseline

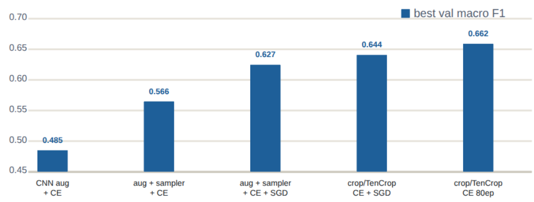

Figure 4: Plot of best training combinations in the baseline model regarding Macro F1 in the validation set.

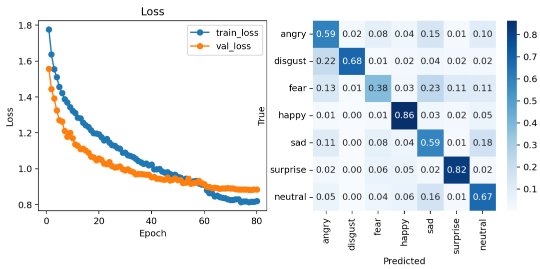

Figure 5: Loss over epochs in the training process and confusion matrix containing the confusion values for the validation dataset. All these results are extracted from the final best combination from baseline model experiments.

Metric	Angry	Disgust	Fear	Happy	Sad	Surprise	Neutral
Precision	0.657	0.67	0.57	0.87	0.53	0.77	0.61
Recall	0.59	0.68	0.38	0.86	0.59	0.82	0.67
F1-score	0.59	0.68	0.45	0.87	0.55	0.79	0.64

Table 5: Per-class performance metrics for the best-performing baseline model on the test set.

6.2 ResNet18

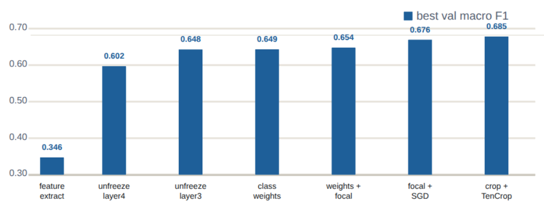

Figure 6: Plot of best training combinations in ResNet18 regarding Macro F1 in the validation set.

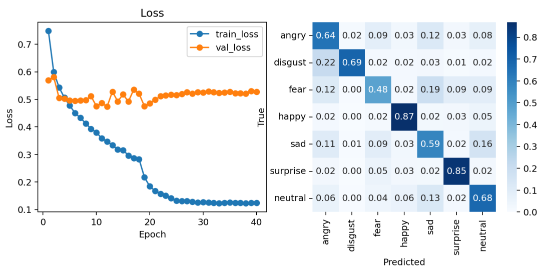

Figure 7: Loss over epochs in the training process and confusion matrix containing the confusion values for the validation dataset. All these results are extracted from the final best combination from ResNet18 experiments.

Metric	Angry	Disgust	Fear	Happy	Sad	Surprise	Neutral
Precision	0.60	0.69	0.60	0.90	0.58	0.77	0.64
Recall	0.64	0.69	0.48	0.87	0.59	0.85	0.68
F1-score	0.62	0.69	0.53	0.88	0.58	0.81	0.66

Table 6: Per-class performance metrics for the best ResNet18 model on the test set.

6.3 EfficientNet-B0

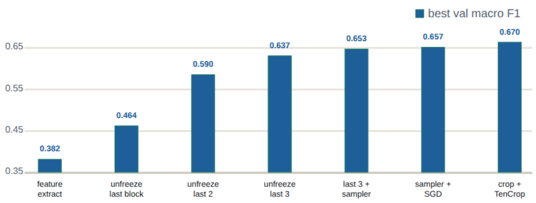

Figure 8: Plot of best training combinations in EfficientNet-B0 regarding Macro F1 in the validation set.

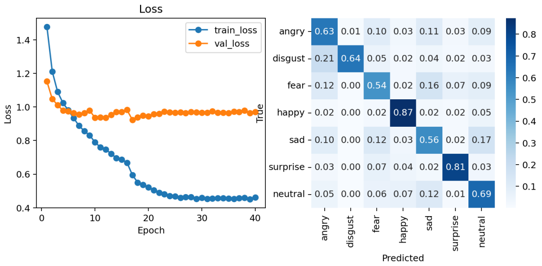

Figure 9: Loss over epochs in the training process and confusion matrix containing the confusion values for the validation dataset. All these results are extracted from the final best combination from EfficientNet-B0 experiments.

Metric	Angry	Disgust	Fear	Happy	Sad	Surprise	Neutral
Precision	0.62	0.77	0.56	0.88	0.59	0.81	0.63
Recall	0.63	0.64	0.54	0.87	0.56	0.81	0.69
F1-score	0.62	0.70	0.55	0.88	0.58	0.81	0.65

Table 7: Per-class performance metrics for the best EfficientNet-B0 model on the test set.

6.4 Comparisons

Model	Weighted F1	Macro F1
Baseline	0.68	0.66
ResNet18	0.70	0.69
EfficientNet-B0	0.69	0.67

Table 8: Comparison of performance between the three best trained models regarding weighted and macro F1 in the test set.

Model	Top Confusion	Count	Rate (%)
Custom CNN	sad → neutral	259	20.8
	fear → sad	233	22.8
	neutral → sad	162	13.1
ResNet18	sad → neutral	200	16.0
	fear → sad	197	19.2
	neutral → sad	162	13.1
EfficientNet-B0	sad → neutral	206	16.5
	fear → sad	163	15.9
	sad → fear	154	12.3

Table 9: Most frequent misclassification patterns across different models on FER2013. The table reports the top confusion pairs, their absolute counts, and relative frequency.

Emotion	ResNet18 [Lit.]	ResNet18	EfficientNet-B0 [Lit.]	EfficientNet-B0	Baseline
Neutral	0.57	0.66	0.70	0.65	0.64
Happy	0.48	0.88	0.91	0.88	0.87
Sad	0.52	0.58	0.58	0.58	0.55
Surprise	0.79	0.81	0.84	0.81	0.79
Fear	0.50	0.53	0.58	0.55	0.45
Disgust	0.68	0.69	0.78	0.70	0.68
Angry	0.57	0.62	0.65	0.62	0.59
Macro-avg	0.64	0.69	0.72	0.67	0.66

Table 10: Comparison of per-class F1-scores and macro-average F1 between literature results and the models evaluated in this work. The literature results are extracted from Helaly et al. [3] and Sufiuh et al. [4]

Emotion	Custom CNN	ResNet18	EfficientNet-B0
Angry	0.48	0.57	0.42
Disgust	0.71	0.66	0.28
Fear	0.43	0.41	0.26
Happy	0.97	0.96	0.96
Sad	0.49	0.69	0.51
Surprise	0.95	0.94	0.91
Average	0.67	0.71	0.56

Table 11: Per-class F1-scores on the CK+ dataset for Custom CNN, ResNet18, and EfficientNet-B0. The last row shows the macro-average F1-score across emotions.

The best-performing configuration differed slightly across architectures. The custom CNN achieved its highest Macro F1 score using weighted sampling, SGD with momentum, learning-rate decay on plateau, crop augmentation, TenCrop evaluation, and an extended 80-epoch training schedule. For ResNet18, the best results were obtained by fine-tuning from Layer 3, modifying the classifier head, and applying class weighting, focal loss, SGD with momentum, learning-rate decay on plateau, and crop augmentation with TenCrop evaluation. EfficientNet-B0 achieved its highest performance by fine-tuning from MBConv Block 2, using a modified classifier head, weighted sampling, SGD with momentum, learning-rate decay on plateau, and crop augmentation with TenCrop evaluation.

Table 10 compares our best-performing models with previously reported results in the literature. ResNet18 achieved a macro-average F1 score of 0.69, exceeding the reported value of 0.64. In contrast, EfficientNet-B0 reached a macro-average F1 score of 0.67, which remained below the literature benchmark of 0.72. Nevertheless, both transfer-learning models outperformed the custom CNN baseline.

Table 11 presents the cross-dataset evaluation results on CK+. Since our CK+ evaluation did not include the neutral class, the CK+ macro-F1 scores should be interpreted as an external robustness check rather than a direct one-to-one replacement for FER2013 test performance. ResNet18 achieved the highest macro-average F1 score (0.71), followed by the custom CNN (0.67), while EfficientNet-B0 showed a substantial drop in performance (0.56).

7 Discussion

Although the three architectures are substantially different, their final validation Macro F1 scores on FER2013 were relatively close. The custom CNN achieved competitive performance despite being trained from scratch, while the transfer-learning models benefited from pre-trained representations. Interestingly, the optimal configurations differed across architectures, particularly regarding the techniques used to address class imbalance (Figures 4, 6 and 8). These observations suggest that optimization choices and training procedures played an important role in the final performance, regardless of the underlying architecture.

For the custom CNN, the loss curves indicate that training had largely converged by epoch 80 (Figure 5). Both training and validation metrics stabilized, suggesting that extending training further would be unlikely to produce meaningful improvements. Therefore, the final model was trained for a total of 80 epochs.

The transfer-learning models exhibited a different behavior. For both ResNet18 and EfficientNet-B0 (Figures 7 and 9), the training loss continued to decrease, whereas the validation loss decreased initially and then reached a plateau with minor fluctuations or a slight rebound. This suggests that further fitting of the training data was no longer translating into substantial improvements on unseen samples, indicating the onset of mild overfitting. Nevertheless, because model checkpoints were selected according to the best validation Macro F1 score with validation loss as tie breaker, the final saved models were largely protected from later-stage overfitting.

Based on the per-class F1 scores on the test set (Figures 5, 6 and 7), the weakest emotions across the models were generally sad and fear. This behavior cannot be fully explained by class imbalance alone, since sad is the third most represented class and fear also has a relatively large number of samples. In contrast, disgust, the least represented class, achieved comparatively strong F1 scores in all models, usually behind only happy and surprise. This suggests that the imbalance-aware strategies helped reduce majority-class dominance and prevented the minority class from being ignored.

The confusion matrices further support this interpretation. The models did not simply over-predict happy, even though happy is both the most represented and the best-performing class.

Instead, the most frequent errors involved visually ambiguous categories, especially sad, fear, neutral, and angry. These observations suggest that some emotions in FER2013 are intrinsically harder to classify beyond their representation percentage, likely because of overlapping facial cues, label ambiguity, and the low resolution of the images.

To better visualize which emotions were most frequently confused, the top confusion pairs for each model were extracted (Table 9). The table shows that the most recurrent confusion patterns across models involve sad, fear, and neutral expressions. These errors were repeated consistently across the models, with several confusion pairs accounting for approximately 10–20% of the samples in the corresponding true class. This pattern is consistent with the per-class F1 results, where fear and sad were among the weakest-performing emotions.

We compared our results with two similar researches, in which they also used ResNet18 and EfficientNet-B0 with transfer learning techniques for the classification of FER2013 dataset. From this comparison (Table 10), we observe that fear and sadness also have a lower performance than the rest of emotions in their results. It is very clear in the case of literature EfficientNet-B0, where almost all emotions surpass an F1 of 0.7, and sadness and fear both have an F1 of 0.58. This also supports the theory that these classes are more difficult to classify in FER2013.

This indicates that the main limitation was not simply learning the class distribution, but distinguishing subtle and overlapping facial expressions. Sad and neutral expressions can share weak or low-intensity facial cues, while fear may overlap with sad, angry, or surprise depending on the visibility of the eyes and mouth. Since FER2013 images are low-resolution grayscale crops, these subtle differences are often difficult to capture. Therefore, although imbalance-aware training and transfer learning helped avoid a simple majority-class bias, they did not fully solve the fine-grained ambiguity between visually similar emotions.

Our best trained model was ResNet18, which achieved the strongest overall performance among the models trained in this project. Although it did not surpass the EfficientNet-B0 reference reported in the literature, its results were close for most emotions. The largest gap with respect to the reference was observed for disgust, where our model obtained an F1 score of 0.69, approximately 0.09 lower than the reported value. This difference may be partly related to the fact that disgust is the least represented emotion in FER2013, meaning that the imbalance-aware techniques used in this project may not have fully compensated for the limited number of training examples. Nevertheless, an F1 score of 0.69 for the smallest class is still a competitive result and suggests that the model did not ignore this minority emotion.

To check how well our models were able to extract the general features of each emotion, we tested their performance on an external dataset, CK+. This dataset was chosen because it shares most emotion labels with FER2013, and contains grayscale facial-expression images. However, CK+ differs from FER2013 in several important aspects: its images have higher resolution(640x490), the expressions are posed, and the samples contain fewer illumination and occlusion artifacts. For these reasons, CK+ should be easier to classify in terms of visual clarity, but good performance on this dataset would still require the models to have learned meaningful emotion-specific features rather than only FER2013-specific patterns.

The per-class F1 scores on CK+ are reported in Table 11. EfficientNet-B0 showed the weakest cross-dataset behavior: its performance decreased for most classes, with the exception of happy and surprise, which improved compared with FER2013. This suggests that, despite its strong FER2013 performance, the model may have learned features that were less robust to the domain shift between FER2013 and CK+, rather than fully general emotion representations.

In contrast, the custom CNN and ResNet18 achieved slightly higher macro-F1 scores on CK+ than on FER2013. However, the per-class results show that this improvement was not uniform. Happy and surprise improved substantially, while harder or more ambiguous classes such as fear and angry remained weaker or decreased. For ResNet18, sad improved on CK+, but fear still dropped noticeably, showing that the difficulty of some emotions persisted across datasets.

These results are consistent with the behavior observed on FER2013. The easier classes, especially happy and surprise, became even easier on CK+, likely because CK+ contains clearer, posed expressions with fewer visual artifacts. However, the models still struggled to generalize evenly across all emotions. Therefore, even though the custom CNN and ResNet18 obtained higher macro-F1 scores on CK+, their performance remained imbalanced across classes, which limits the claim that they fully extrapolate to external datasets. ResNet18 was the closest to doing so, since it maintained F1 scores above 0.60 for most emotions, but fear remained a clear limitation.

This external validation highlights an important limitation of the project. Strong performance on FER2013 does not necessarily imply robust cross-dataset generalization. The improvement of happy and surprise on CK+ suggests that the models can recognize expressions with clear and distinctive facial cues, especially when the images are cleaner and more controlled. However, the weaker transfer for fear, angry, and disgust shows that the learned representations are less reliable for emotions with subtler or more variable visual patterns. This indicates that the models still depend partly on dataset-specific characteristics, and that training on a single dataset limits their ability to generalize to new acquisition conditions.

8 Conclusions

From all the analysed results, we can conclude that ResNet18 was the best-performing model in this project, as it maintained strong overall performance on both FER2013 and CK+. Nevertheless, it still struggled with some emotions, especially fear, and to some extent sadness and anger.

Transfer learning with ResNet18 and EfficientNet-B0 was also successful on FER2013, since both models were able to classify facial emotions effectively despite being initialized with ImageNet-pretrained weights rather than trained from scratch on FER2013. This supports the idea that general visual features learned from a different dataset, such as edges, textures, and simple shapes, can still be useful for facial expression recognition.

The imbalance-aware techniques seemed to help, since disgust, the least represented emotion in the training dataset, achieved relatively strong performance across the FER2013 models.

We also observed the importance of analysing individual emotion classes, since aggregate scores can hide weak class performance. This was evident in the CK+ evaluation of the custom CNN and ResNet18. Although their macro-F1 scores were slightly higher than on FER2013, the improvement was not uniform across emotions. Happy and surprise improved substantially, while harder classes such as fear and angry remained weak or decreased. Therefore, aggregate metrics alone were not sufficient to evaluate whether the models generalized well across all emotion categories.

External validation also showed that testing models on additional datasets can reveal issues that are not visible from the original test set alone. This was particularly evident for EfficientNet-B0, whose performance decreased substantially when evaluated on CK+. External datasets are therefore useful for detecting dataset-specific overfitting and for assessing whether the features learned by the models transfer to similar but independently collected data.

From the performance on FER2013, CK+, confusion matrices and comparisons with similar works, we observed that some emotions were consistently more difficult to classify than others, specifically sad, fear, neutral and anger. Since these emotions are not underrepresented in FER2013, their difficulty is likely not explained by class imbalance alone. Instead, it may be related to the intrinsic ambiguity of the expressions themselves, or to image-level issues such as occlusions, strong illumination variations, low resolution, and noisy labels.

For future work, we would like to establish a human-performance baseline on a subset of FER2013 and CK+. This would involve manually annotating a representative sample of images and comparing human predictions with both the original dataset labels and the model predictions. Such an analysis would help determine whether the low performance on some emotions is mainly caused by model limitations or by the intrinsic ambiguity of the images themselves.

This human annotation study would also allow us to estimate label noise in FER2013. If human annotators frequently disagree with the original labels, especially for emotions such as fear, sad, neutral, and anger, this would support the hypothesis that some errors are caused by ambiguous or noisy labels rather than by the model alone. In future work, relabeling or filtering these uncertain samples could improve training quality and lead to more reliable model evaluation.

Another possible direction would be to use soft labels for ambiguous samples. Instead of assigning a single hard emotion label to each image, soft labels would represent uncertainty by distributing probability mass across multiple plausible emotions. For example, an ambiguous image could be labeled partly as fear and partly as sadness or surprise. This approach could better reflect the overlap between facial expressions and reduce the penalty for reasonable alternative predictions, potentially improving robustness on visually ambiguous classes.

Finally, future work should extend the external validation beyond CK+ by evaluating the models on additional facial-expression datasets. This would make it possible to test whether the observed behavior is specific to CK+ or reflects a broader generalization limitation. In addition, stronger pretrained backbones or ensemble approaches could be explored to improve robustness. For example, combining models with different architectures could help reduce model-specific errors and provide more stable predictions across ambiguous emotion classes.

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